Voltage Control Strategies for Distribution Networks with Distributed Energy Resources

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Abstract — This publication presents a research on the Voltage Control Strategies for Distribution networks with Distributed Energy Resources (DER). Taking into account the network topology and the load and generation profiles some typical states are calculated using the STATUS power system analysis software. Furthermore, a quasi-dynamic analysis and test case with distributed photovoltaic generators (PVG) in a distribution network is performed in order to demonstrate the interaction of different converter interfaced DER and other network controlled elements such as under load tap changers. Based on the results obtained valuable conclusions and recommendations are made.

Keywords— voltage control, distribution network, distributed energy resources, smart grid

I. Introduction

With the increased share of DER in Electric Power System (EPS) in the recent years new challenges for the control of distribution network's parameters especially the voltage arise [1]. Which control strategies should be used, in which cases? And is there a best one? These questions constantly reappear in different forms [1-27].

The study examines several different voltage control strategies, means and options. A simulation of an existing distribution network with DER is performed to respond some of the questions posed.

II. Voltage Control Methodology, Strategies, Approaches and Means

In this study several applicable voltage control strategies will be considered and shortly presented below.

A. Under load tap changer (ULTC)

ULTC is an essential part of a power transformer. It adjusts the output voltage of the transformer by changing the tap position to respond to changes in power generation and consumption [1]. The ULTC can automatically change the taps, adjust the transformation ratio, adjust the voltage during the system operation, which makes it an important voltage control equipment in the power system. It has been widely used in high voltage (HV)/middle voltage (MV) transformers and recently it is also considered as promising voltage control solution for MV/low voltage (LV) transformers in active networks with distributed generation [1], **Error! Reference source not found.**. Fig. 1 presents an example of ULTC operation.

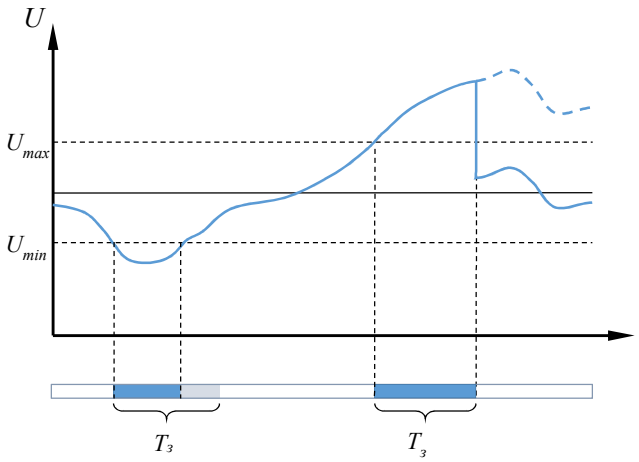


Fig. 1. Voltage control using ULTC

The voltages U_{min} and U_{max} define the boundaries in which the voltage can change. If the voltage goes beyond the boundaries set, the ULTC switches the tap after a specified time delay. This operation changes the transformer's ratio which brings back the voltage within the boundaries.

B. Active Power Limitation (APL)

The APL strategy is obligatory and widely implemented in solar inverters **Error! Reference source not found.**. Under normal conditions the PVG injects its maximum possible generated power P_{max} to the network. If the voltage reaches specified threshold U_{min}, the inverter starts to limit the active power injected to the network until it reaches zero at U_{max}. In this way the inverter controls and limits the voltage rise at the point of common coupling (PCC) when it is caused by himself (Fig. 2).

C. Voltage control using Reactive Power Control (RPC)

By consuming or generating reactive power the generator can influence the network power flow which in turn changes the

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voltage at the PCC. The voltage control method using reactive power has several variations:

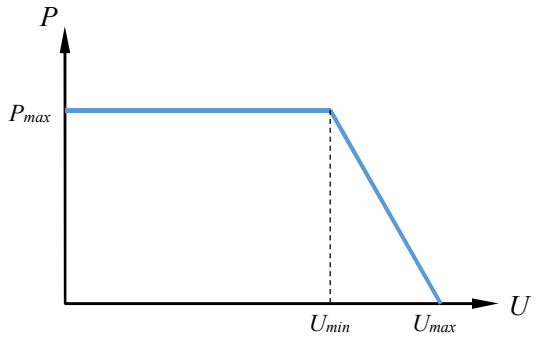


Fig. 2. Voltage control using APL

1) Constant power factor (cos(ф)=const): in this approach during the voltage control process the power factor is kept constant (Fig. 3).

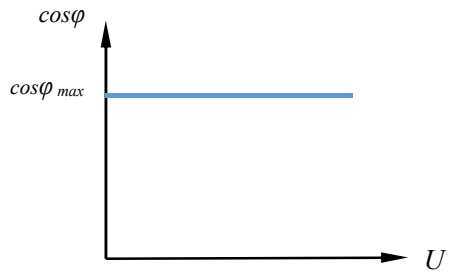


Fig. 3. Voltage control with constant power factor

2) Power factor dependent on active power (cos(ф)=f(P)): in this approach during the voltage control process the power factor is dependent on the active power. With P=0 cos(ф)=cos(ф)max, and with P=Pmax cos(ф)=cos(ф)min respectively *(Fig. 4).*

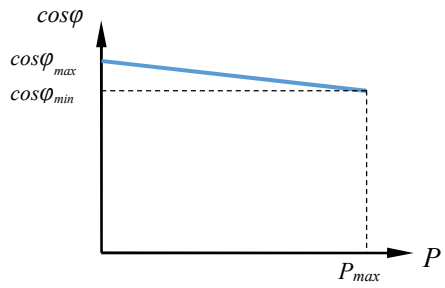


Fig. 4. Voltage control with power factor dependent on active power

3) Voltage control with power factor dependent on voltage (cos (ф)=f(U)): in this approach during the voltage control process the power factor is dependent on the voltage. With $U=U_{min}$, $cos(ф)=cos(ф)_{max}$, respectively with $U=U_{max}$, $cos(ф)=cos(ф)_{min}$ (Fig. 5).

4) Voltage control with reactive power dependent on voltage (Q=f(U)): in this approach reactive power is consumed or generated within the corresponding PQ diagram (Fig. 6) in order to keep the voltage as close as possible to the reference value $U_{ref.}$ (Fig. 7). In case of reaching the limits of the generators PQ diagram (Fig. 6) the method loses its controllability and the reactive power is set to its minimum (Q_{min}) or maximum (Q_{max}) boundary value allowed.

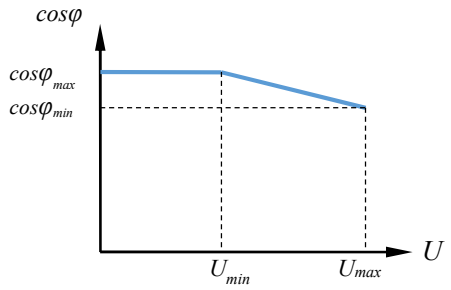


Fig. 5. Power factor dependent on voltage

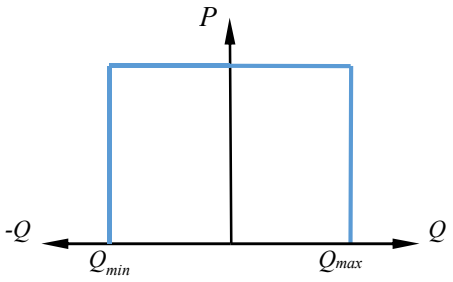


Fig. 6. PQ diagram of typical PV generator

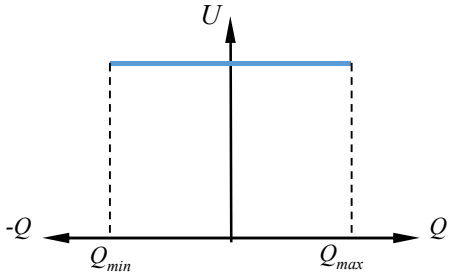


Fig. 7. Voltage control using reactive power

III. Test System Description

In order to study and demonstrate the performance of the voltage control strategies an existing real life MV+LV distribution network with 832 nodes and 832 branches is considered. The network is hosting 26 PVG with total installed capacity of 3251 kWp which are distributed in both 0,4 kV LV network (Fig. 8) and 20 kV MV network (Fig. 9) **Error! Reference source not found.**.

The LV network consists of 646 branches, 645 nodes and 2 transformer stations (TS1 and TS2) with installed capacity of 400KVA each. Feeding nearly 524 kW predominantly domestic load of the village the network hosts 10 distributed PVGs with approximate unit installed capacity of 30kWp totalling 289.5 kWp.

The MV network consists of 186 branches, 187 nodes and hosts another 16 PVGs with total installed power of 2962 kWp.

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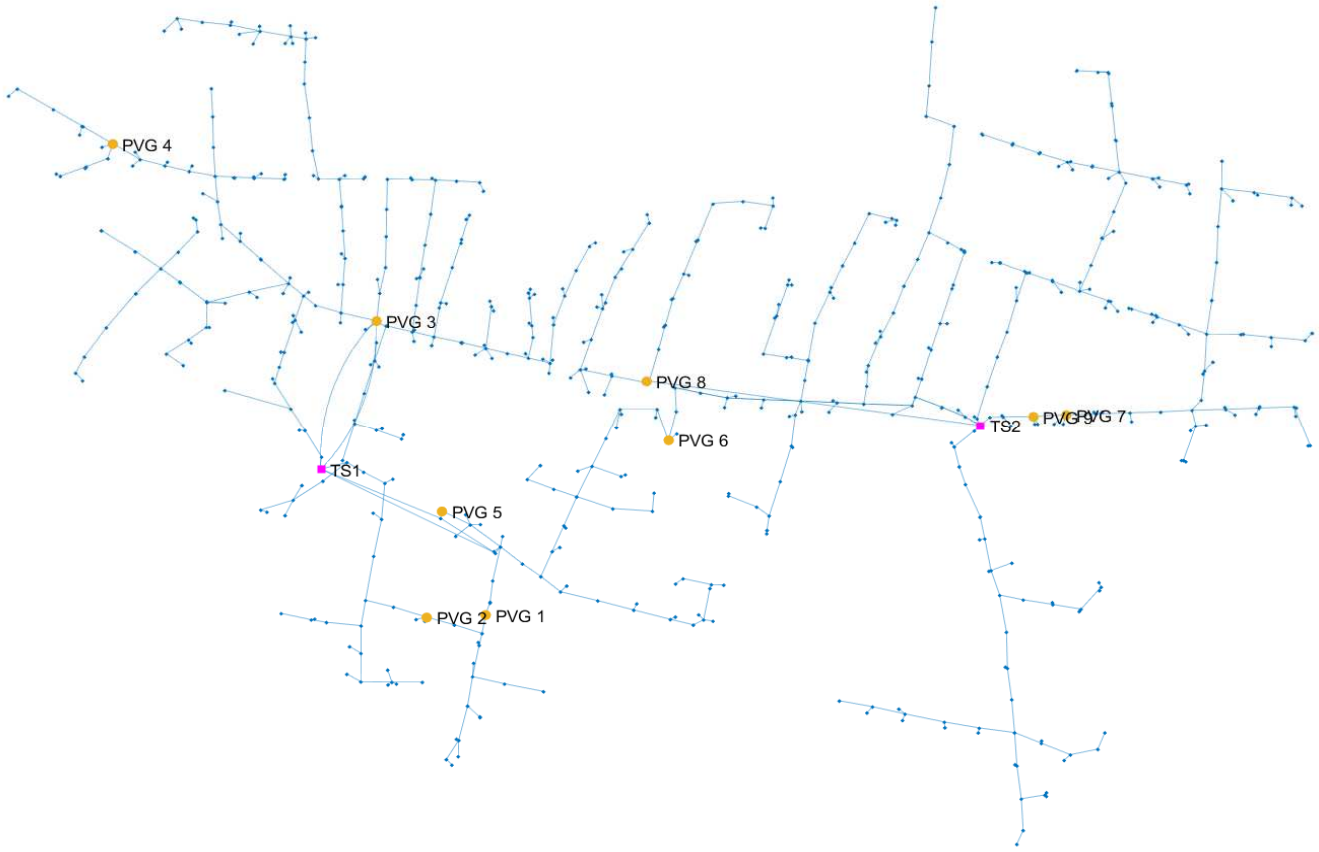


Fig. 8. Low voltage network

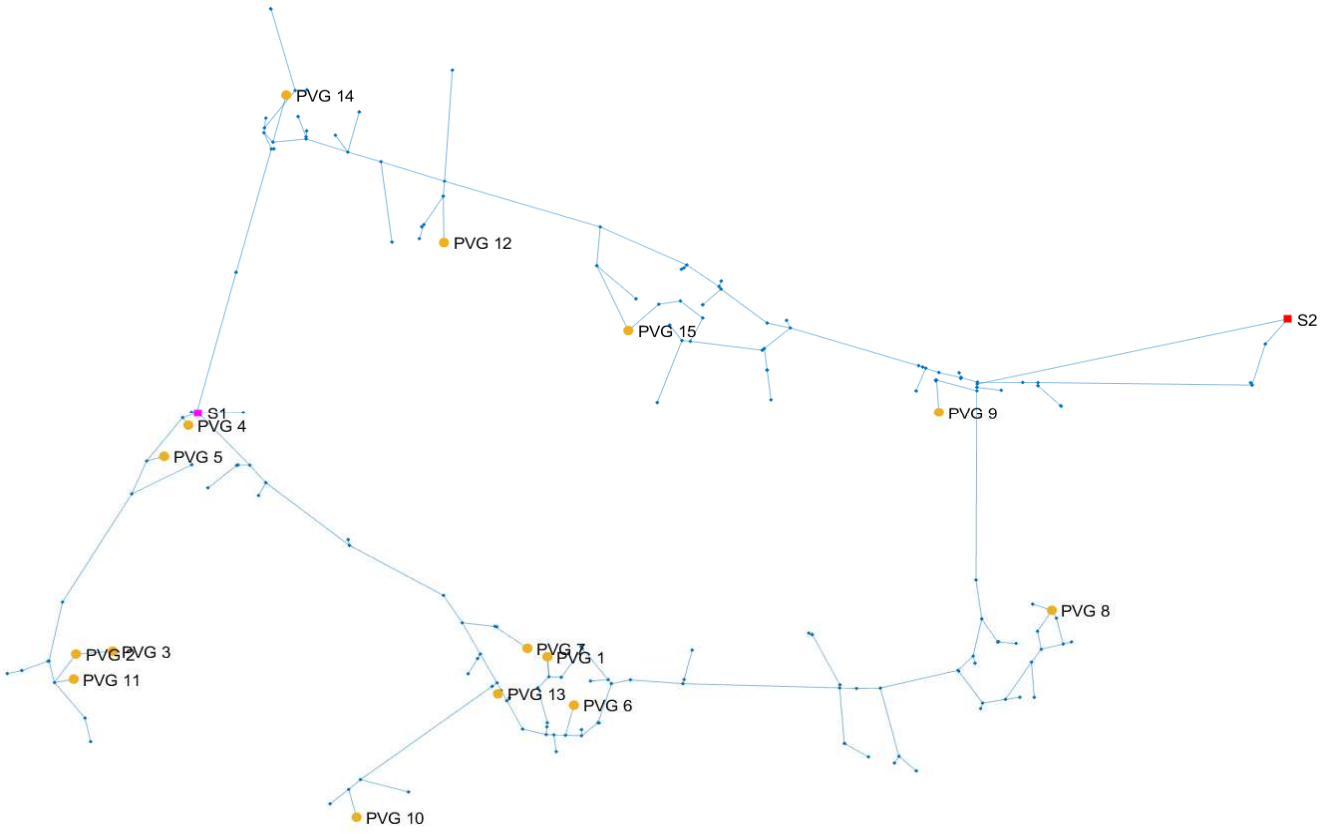


Fig. 9. Middle voltage network

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IV. Modeling, Analysis and Validation

A quasi dynamic analysis using the STATUS software is performed [2], [5], [26].

To validate the network model and evaluate the calculation accuracy a comparative analysis for a representative typical operation state using STATUS, NEPLAN and DMS software is performed.

As a first step validation between STATUS and NEPLAN is conducted on the MV network model with average power factor of 0.87 and slack bus voltage 20kV. The maximum deviation between the software tools is 0.0585%.

As a second step validation between STATUS, NEPLAN and DMS on the same network model is conducted with power factor of 0.95 and slack bus voltage of 20.6 kV. The results show 0.05% deviation between STATUS and DMS and 0.00035% between STATUS and NEPLAN **Error! Reference source not found.**.

The validation results confirm that the deviation between STATUS, NEPLAN and DMS is within admissible accuracy boundaries set.

To better focus on the voltage control performance all nodal loads are excluded and only the time varying generations at each PVG node are considered.

V. Results

The complete study goes beyond the purpose of this publication and therefore only the most distinguished cases will be presented below.

A. Under load tap changer

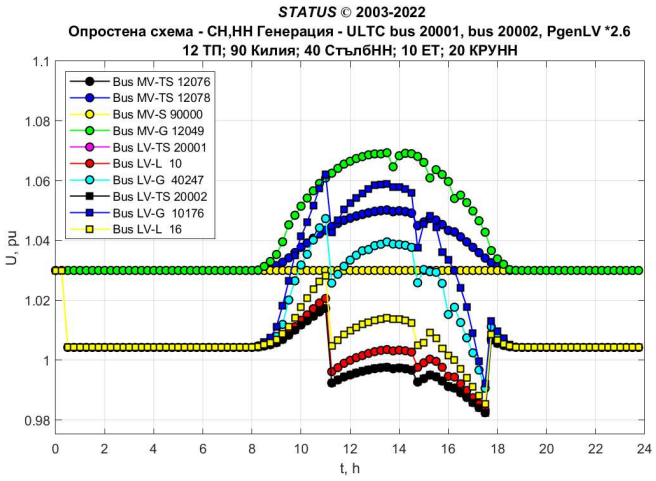


Fig. 10. ULTC - Voltage

TABLE I. ULTC RESULTS

Network	Umin, pu	Umax, pu	ΔU,%	Eg,kWh	ΔE, kWh	ΔE,%
MV	1,030	1,069	3,938	13842,65	426,10	3,08
LV TS1	0,982	1,047	6,493	1310,82	24,97	1,90
LV TS2	0,982	1,062	7,979	1529,45	38,76	2,53
LV Total	0,982	1,062	7,982	2840,27	63,73	2,24

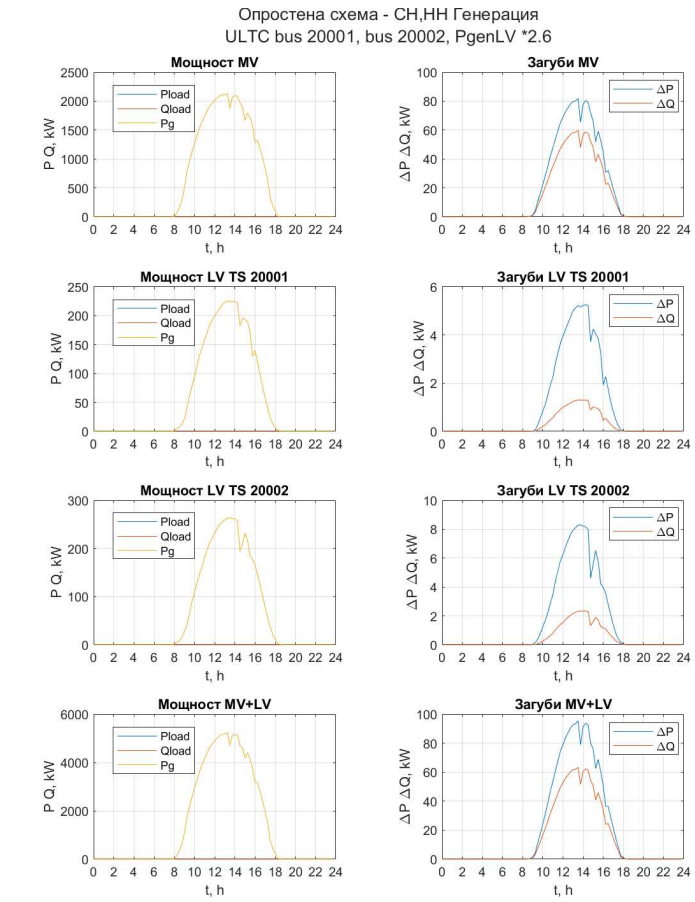


Fig. 11. ULTC – Power flow and Power losses

B. Active Power Limitation

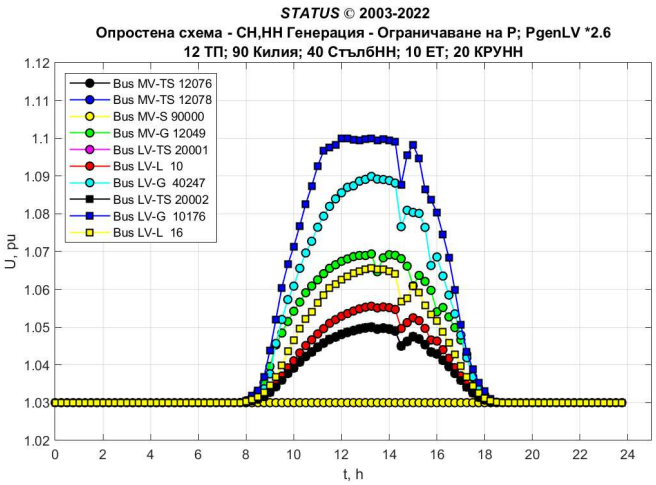


Fig. 12. APL – Voltage

TABLE II. APL RESULTS

Network	Umin, pu	Umax, pu	ΔU,%	Eg,kWh	ΔE, kWh	ΔE,%
MV	1,03	1,069	3,938	13842,65	425,56	3,07
LV TS1	1,03	1,090	5,990	1310,82	22,76	1,74
LV TS2	1,03	1,100	6,998	1507,89	33,36	2,21

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Network	Umin, pu	Umax, pu	ΔU,%	Eg,kWh	ΔE, kWh	ΔE,%
LV Total	1,03	1,100	6,998	2818,72	56,12	1,99

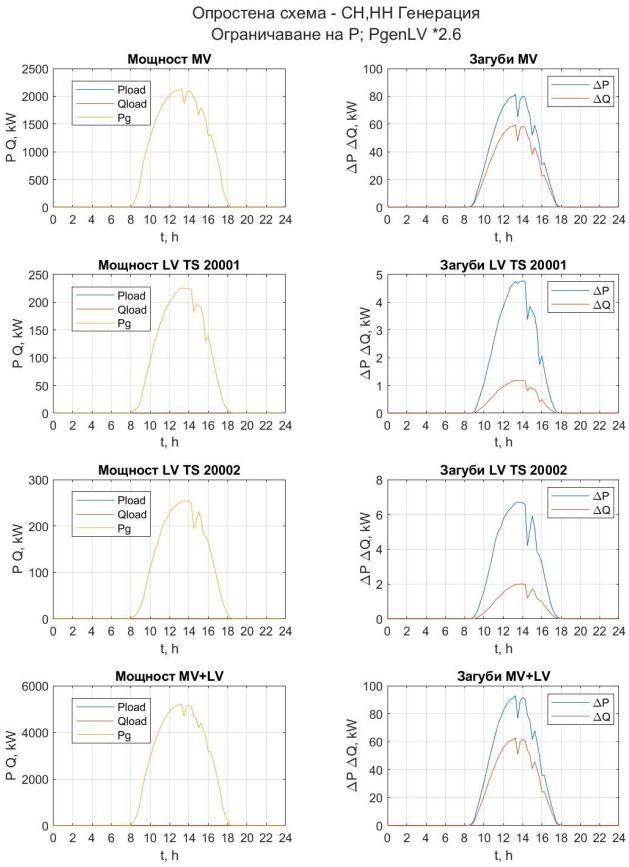


Fig. 13. APL – Power flow and Power losses

C. *Reactive Power Control*

1) *Constant power factor*

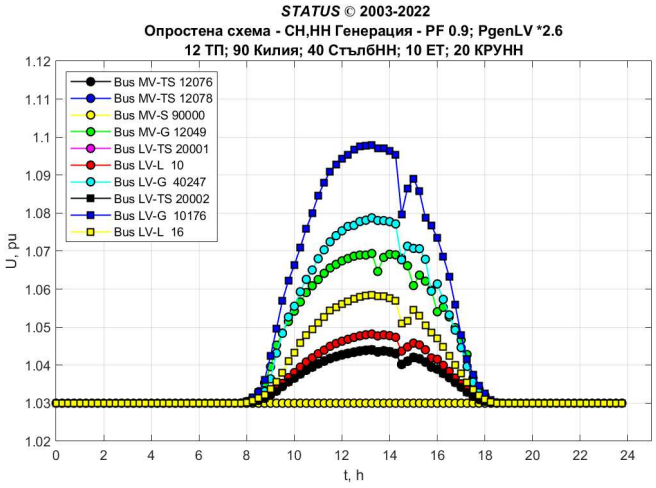


Fig. 14. RPC *cosф=const*

TABLE III. RPC RESULTS

Network	Umin, pu	Umax, pu	ΔU,%	Eg,kWh	ΔE, kWh	ΔE,%
MV	1,030	1,069	3,938	13842,65	434,27	3,14

Network	Umin, pu	Umax, pu	ΔU,%	Eg,kWh	ΔE, kWh	ΔE,%
LV TS1	1,030	1,079	4,876	1310,82	28,53	1,55
LV TS2	1,030	1,098	6,798	1529,45	44,38	2,04
LV Total	1,030	1,098	6,798	2840,27	72,91	2,57

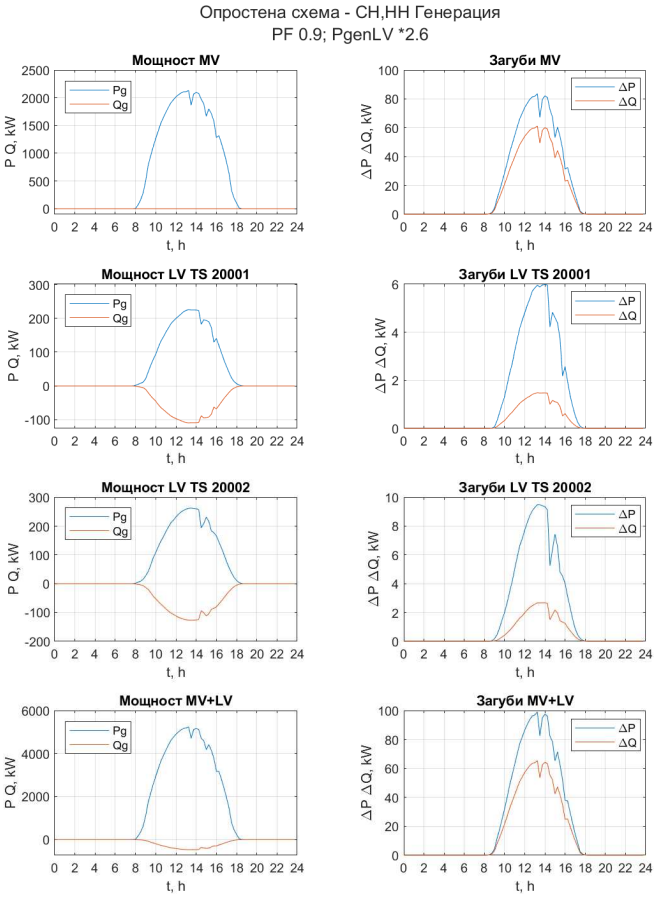


Fig. 15. RPC *cosф=const* - Power flow and Power losses

2) *Power factor dependent on active power*

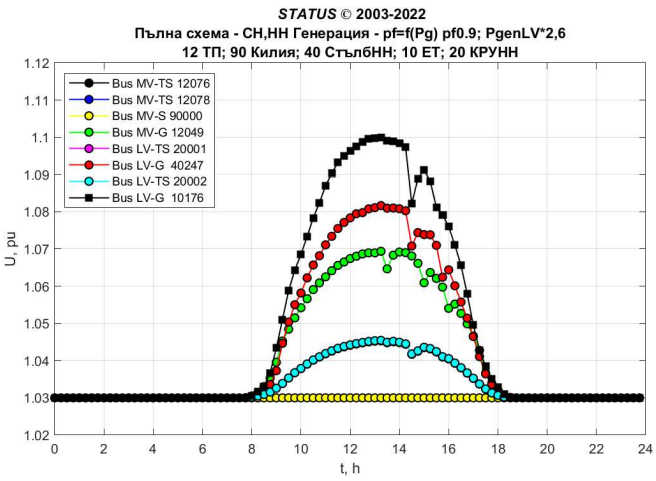


Fig. 16. RPC *cosф=f(P)*

TABLE IV. RPC RESULTS

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Network	Umin, pu	Umax, pu	ΔU,%	Eg,kWh	ΔE, kWh	ΔE,%
MV	1,030	1,069	3,938	13842,65	430,69	3,11
LV TS1	1,030	1,082	5,166	1310,82	21,27	1,33
LV TS2	1,030	1,100	6,997	1529,45	40,57	2,17
LV Total	1,030	1,100	6,997	2840,27	61,84	2,18

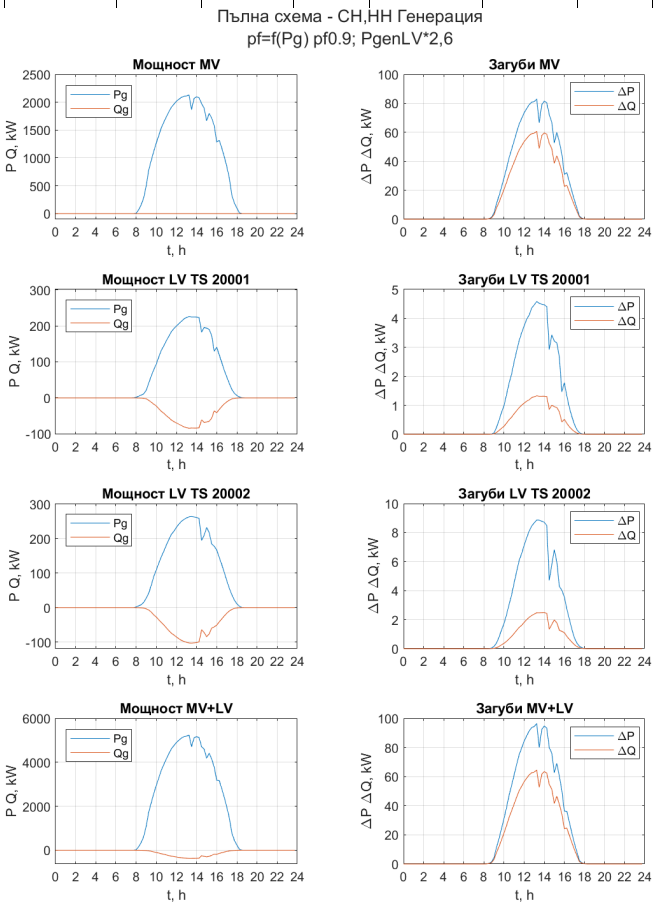


Fig. 17. RPC *cosφ=f(P)* - Power flow and Power losses

Network	Umin, pu	Umax, pu	ΔU,%	Eg,kWh	ΔE, kWh	ΔE,%
MV	1,030	1,069	3,938	13842,65	431,70	3,12
LV TS1	1,030	1,081	5,069	1310,82	21,96	1,32
LV TS2	1,030	1,099	6,865	1529,45	42,49	2,15
LV Total	1,030	1,099	6,865	2840,27	64,45	2,27

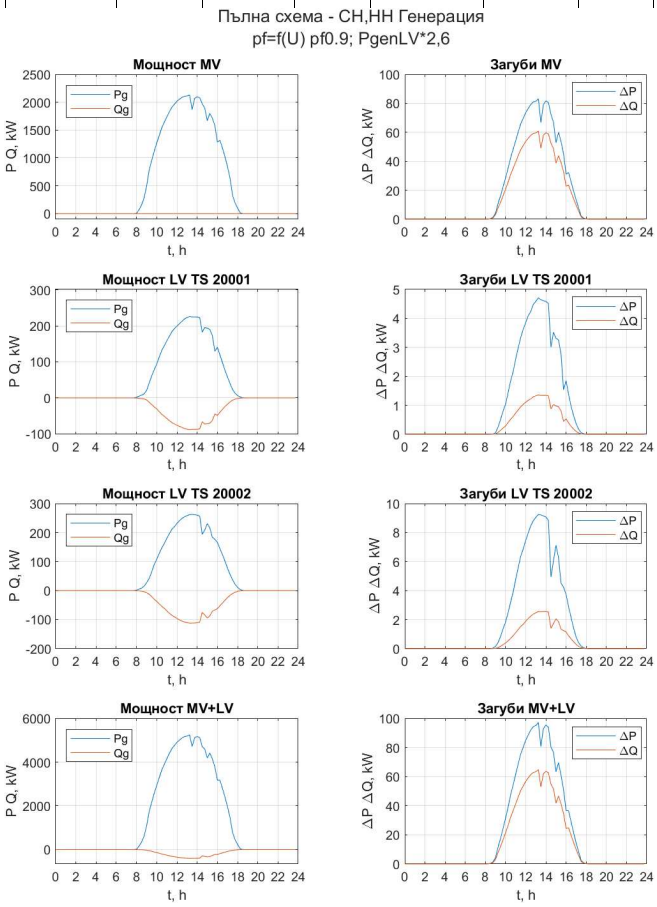


Fig. 19. RPC *cosφ=f(U)* - Power flow and Power losses

3) *Power factor dependent on voltage*

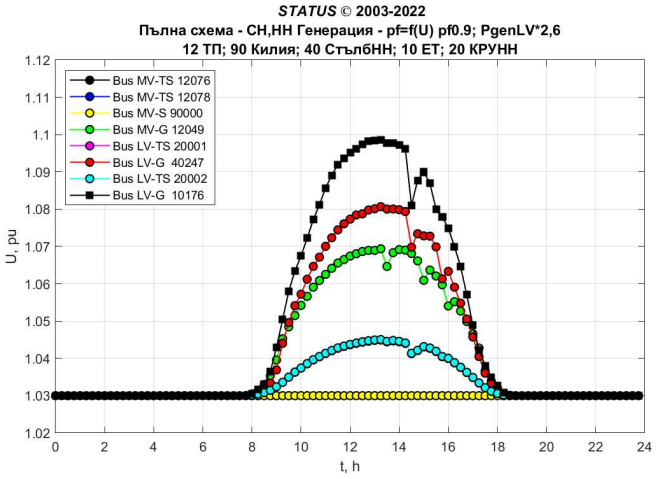


Fig. 18. RPC *cosφ=f(U)*

TABLE V. RPC RESULTS

4) *Reactive power dependent on voltage*

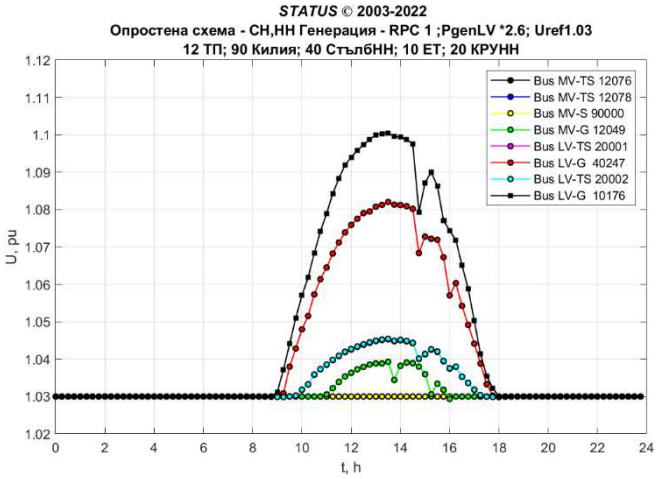
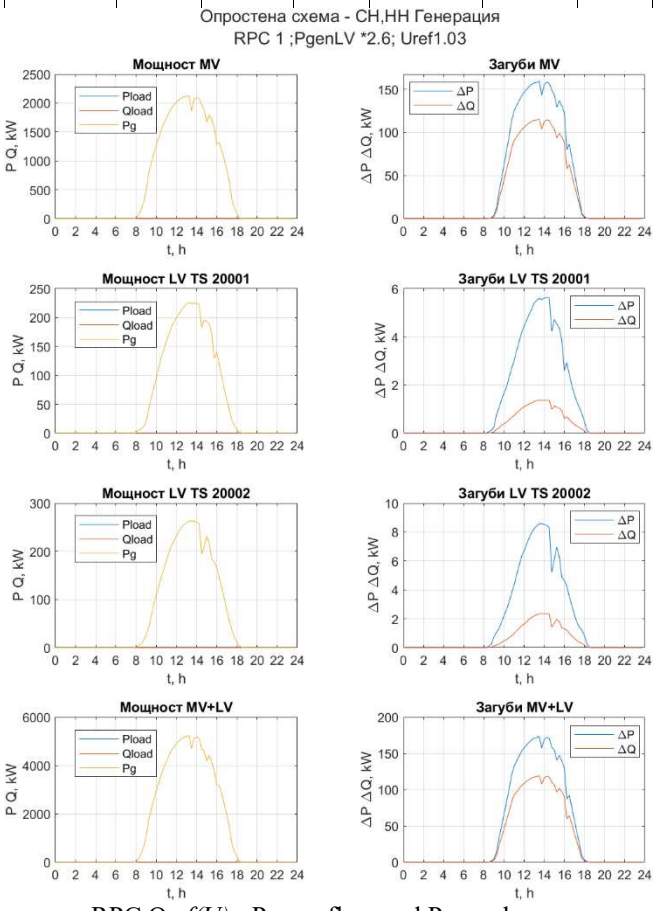


Fig. 20. RPC Q=*f(U)*

TABLE VI. RPC RESULTS

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Network	Umin, pu	Umax, pu	ΔU,%	Eg,kWh	ΔE, kWh	ΔE,%
MV	1,029	1,045	1,628	13842,65	960,66	6,94
LV TS1	1,030	1,082	5,244	1310,82	30,46	2,32
LV TS2	1,030	1,101	7,072	1529,45	44,92	2,94
LV Total	1,030	1,101	7,093	2840,27	75,38	2,65



RPC Q=*f(U)* - Power flow and Power losses

VI. Conlusion

The ULTC control strategy is proving to be very effective mean for voltage control. It allows increased power generation and in same time it has some drawbacks such as need of significant investments, higher maintenance cost and lower reliability.

The APL is a simple, very effective and reliable solution applied directly to the PVG. It is essential and mandatory voltage control option which prevents from abnormal operation with inadmissibly high voltages. It does not require any additional investments or maintenance costs. In specific situations when the PVG's injected power is restricted to some extent it might cause some drawbacks. Moreover, if additional control is not applied, the amount of restricted power and energy will be different depending on the position of each of the different PVGs in the network. Another important issue which needs to be considered is the time of the maximum PV production appearance, especially considering how this influences the others producers and network users. The duration of time in which APL is activated and the amount of limited active power and energy which specific PVG cannot supply to the network may not always lead to considerable losses. Thus, it might be acceptable solution in many cases. However, the APL strategy does not allow infinite integration of PVG into the network and also it does not repeal the necessity of evaluation of the available maximum possible network transfer capacity.

As in the case of APC, RPC is also applied directly at the PCC of the PVG and does not require additional investments or maintenance costs. It appears to be effective but not always efficient solution. Due to the additional reactive power flows, considerably higher network transfer losses might arise. It should be used with caution, especially in cases where the benefit from the additionally produced energy might not cover the losses and penalties for low power factor. Therefore additional evaluation and cost-benefit analysis should be conducted for each specific case.

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